

Tom Deng Software Engineer

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Technical Skills

Programming Languages: JavaScript, Python, Java, SQL, C++, C, PHP, HTML/CSS

Frameworks & Libraries: Node.js, React, Express.js

Backend & APIs: RESTful API design, OAuth 2.0/single sign-on, WebSockets / real-time communication, RabbitMQ

Databases: PostgreSQL, MongoDB, MySQL

Developer Tools: Git, Docker, VS Code, IntelliJ, PyCharm, Visual Studio, Eclipse, WebSockets

API Technologies: OAuth 2.0, RESTful APIs

Education

University of Waterloo B. Sc. in Computer Engineering

Waterloo, Ontario Apr. 2025

Work Experience

Accion Labs **Software Engineering Intern**

Toronto, Ontario Jan. 2024 - Apr. 2024

- Built a fully responsive timesheet application interface in **React**, **Redux Toolkit** and **Tailwind CSS**, enabling over **100** employees to track work hours, and applied **Client-Side Rendering** to achieve page loads under **300 milliseconds**.
- Designed and implemented secure backend services in **Node.js** with **PostgreSQL**, integrated with **Auth0**, following the Repository Pattern within a Microservices architecture to ensure scalability and maintainability.
- Packaged the application as a **Chrome Extension** using **Manifest V3**, **Chrome Extensions API** and **React**, with **Continuous Deployment** pipelines that reduced deployment times to under **10 minutes**.
- Built an automated testing suite in **Jest** and **Cypress**, integrated with **GitHub Actions**, ensuring end-to-end reliability and achieving **85%** code coverage.
- Configured scalable cloud deployment environments with **Docker** and **Kubernetes**, reducing environment inconsistency errors.

Synqli Inc. **Software Engineering Intern**

Waterloo, Ontario Jan. 2021 - Apr. 2021

- Built a high-performance collaborative filtering recommendation system in **Python** using **Surprise**, **Spark** and **Ray**, achieving prediction latency under **250 milliseconds** while improving scalability and accuracy.
- Integrated **MySQL** as the primary data store, applying the **Repository Pattern**, **indexing strategies** and **connection pooling** to optimise query efficiency.
- Resolved production bugs and handled edge cases in a live user-facing environment, reducing **mean time to resolution**.

Canadian Mortgages Inc. **Software Engineering Intern**

Mississauga, Ontario Sep. 2019 - Dec. 2019

- Designed and implemented an email-queue system using **Apache Kafka** and **PostgreSQL** to handle scheduling, prioritisation, retries and deletions, maintaining processing latency under **50 milliseconds**.
- Enhanced user engagement by **40 percent** through **WebSocket**-based notifications implemented with **Socket.IO**, **WebSocket API** and **Apache Kafka**.
- Automated **PDF** compression workflows using **Ghostscript** via **Apache Airflow**, reducing memory usage by **30 percent**.
- Created and ran automated tests for public **API** endpoints using **Postman** and **pytest**, lowering defect escape rates.

University of Waterloo **Software Engineering Intern**

Waterloo, Ontario Jan. 2019 - Apr. 2019

- Improved modularity and long-term maintainability of a legacy codebase by integrating a micro-framework based on the **MVC** paradigm using **Flask** and **Jinja2**, and applying incremental refactoring.
- Improved frontend components using **ARIA** and **Axe-Core**, applying Server-Side Rendering to meet **WCAG 2.0** standards and restructuring and optimizing codebases for static analysis.